## 14-2b The Marginal-Cost Curve and the Firm’s Supply Decision

To extend this analysis of profit maximization, consider the cost curves in Figure 1. These cost curves exhibit the three features that, as we discussed in the previous chapter, are thought to describe most firms: The marginal-cost curve ($MC$) slopes upward, the average-total-cost curve ($ATC$) is U-shaped, and the marginal-cost curve crosses the average-total-cost curve at the minimum of average total cost. The figure also shows a horizontal line at the market price ($P$). The price line is horizontal because a competitive firm is a price taker: The price of the firm’s output is the same regardless of the quantity that the firm produces. Keep in mind that, for a competitive firm, the price equals both the firm’s average revenue ($AR$) and its marginal revenue ($MR$).

**Figure 1**

**Profit Maximization for a Competitive Firm**

This figure shows the marginal-cost curve ($MC$), the average-total-cost curve ($ATC$), and the average-variable-cost curve ($AVC$). It also shows the market price ($P$), which for a competitive firm equals both marginal revenue ($MR$) and average revenue ($AR$). At the quantity $Q_1$, marginal revenue $MR_1$ exceeds marginal cost $MC_1$, so raising production increases profit. At the quantity $Q_2$, marginal cost $MC_2$ is above marginal revenue $MR_2$, so reducing production increases profit. The profit-maximizing quantity $Q_{\text{MAX}}$ is found where the horizontal line representing the price intersects the marginal-cost curve.

We can use Figure 1 to find the quantity of output that maximizes profit. Imagine that the firm is producing at $Q_1$. At this level of output, the marginal-revenue curve is above the marginal-cost curve, indicating that marginal revenue is greater than marginal cost. This means that if the firm were to raise production by 1 unit, the additional revenue $(MR_1)$ would exceed the additional cost $(MC_1)$. Profit, which equals total revenue minus total cost, would increase. Hence, if marginal revenue is greater than marginal cost, as it is at $Q_1$, the firm can increase profit by increasing production.

A similar argument applies when output is at $Q_2$. In this case, the marginal-cost curve is above the marginal-revenue curve, showing that marginal cost is greater than marginal revenue. If the firm were to reduce production by 1 unit, the costs saved $(MC_2)$ would exceed the revenue lost $(MR_2)$. Therefore, if marginal cost is greater than marginal revenue, as it is at $Q_2$, the firm can increase profit by reducing production.

Where do these marginal adjustments to production end? Regardless of whether the firm begins with production at a low level (such as $Q_1$) or at a high level (such as $Q_2$), the firm will eventually adjust production until the quantity produced reaches the profit-maximizing quantity $Q_{\text{MAX}}$. This analysis yields three general rules for profit maximization:

- If marginal revenue is greater than marginal cost, the firm should increase its output.
- If marginal cost is greater than marginal revenue, the firm should decrease its output.
- At the profit-maximizing level of output, marginal revenue equals marginal cost.

These rules are the key to rational decision making by any profit-maximizing firm. They apply not only to competitive firms but, as we will see in the next chapter, to other types of firms as well.

We can now see how the competitive firm decides what quantity of its good to supply to the market. Because a competitive firm is a price taker, its marginal revenue equals the market price. For any given price, the competitive firm's profit-maximizing quantity of output is found by looking at the intersection of the price with the marginal-cost curve. In Figure 1, that quantity of output is $Q_{\text{MAX}}$.

Suppose that the price prevailing in this market rises, perhaps because of an increase in market demand. Figure 2 shows how a competitive firm responds to the price increase. When the price is $P_1$, the firm produces quantity $Q_1$, the quantity that equates marginal cost to the price. When the price rises to $P_2$, the firm finds that marginal revenue is now higher than marginal cost at the previous level of output, so the firm increases production. The new profit-maximizing quantity is $Q_2$, at which marginal cost equals the new, higher price. *In essence, because the firm's marginal-cost curve determines the quantity of the good the firm is willing to supply at any price, the marginal-cost curve is also the competitive firm's supply curve.* There are, however, some caveats to this conclusion, which we examine next.

**Figure 2**

### Marginal Cost as the Competitive Firm's Supply Curve

An increase in the price from $P_1$ to $P_2$ leads to an increase in the firm's profit-maximizing quantity from $Q_1$ to $Q_2$. Because the marginal-cost curve shows the quantity supplied by the firm at any given price, it is the firm's supply curve.